

EXPERIMENT 6a – DATA VISUALIZATION USING POWER BI

Aim

To load the tourism dataset into Power BI, transform the data, and visualize it to obtain useful insights about tourist locations, visitors, ratings, and revenue.

Algorithm: Tourism Dashboard Calculation

Step 1: Input Data Loading

- Read the tourism dataset containing the columns:
 - Location
 - Country
 - Category
 - Visitors
 - Rating
 - Revenue
 - Accommodation_Available
- Load the dataset into Power BI.

Step 2: Data Transformation

- Check the dataset for missing or incorrect values.
- Verify the data types of Visitors, Rating, and Revenue.
- Prepare the dataset for visualization.
- Apply filters based on Country and Category.

Step 3: KPI Calculation

- Calculate the total number of visitors.

Total Visitors = Sum of Visitors column

- Calculate the average rating.

Average Rating = Average of Rating column

- Calculate the total revenue.

Total Revenue = Sum of Revenue column

Display these values using KPI Cards.

Step 4: Bar Chart Calculation

Tourist Locations by Visitors

- Group the tourism data based on Location.
- Calculate the total number of visitors for each location.
- Sort the locations in descending order based on the number of visitors.
- Display the top tourist locations using a Bar Chart.

Step 5: Donut Chart Calculation

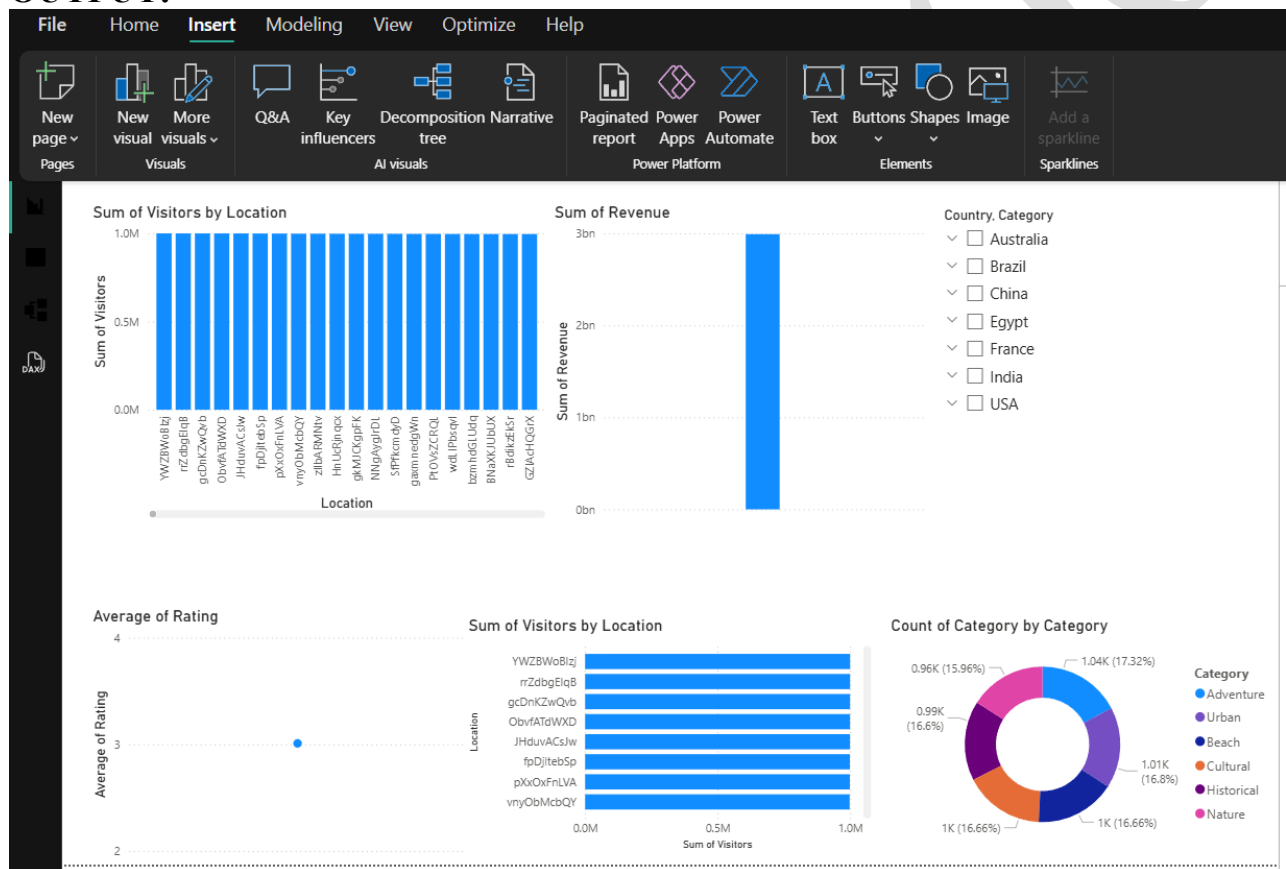
Tourism Distribution by Category

- Group the data based on Category.
- Calculate the number of tourist locations in each category.

Step 6: Dashboard Render

- Display KPI Cards for:
 - Total Visitors
 - Average Rating
 - Total Revenue
- Display a Bar Chart showing tourist locations based on Visitors.
- Display a Donut Chart showing tourism distribution by Category.
- Add slicers for:
 - Country
 - Category
- When the user selects a Country or Category, update all charts and KPI values accordingly.

OUTPUT:



RESULT: